

Cross-Platform Gaming, Auth & Economy Master Ecosystem Manual

Comprehensive blueprint covering Google Firebase Auth, Google Cloud Firestore atomic economy, Google Play Production lifecycle, Meta Facebook OAuth, Next.js legal compliance, and Android Studio SDK tooling.

■ Google Firebase & Firestore Google Sign-In, Email/Password, Facebook Auth, and Cloud Firestore 7-tier Universal Wallet.	■ 7-Tier Universal Economy Spider Brass, Copper, Silver, Gold, Diamond, Jade, & Obsidian coin balances synced across all titles.
■ Google Play & BillDesk IPV Version 1.2.1 Build 9 Live on Production track with verified Indian merchant payments profile.	■ Official Web & Compliance webspiderstudios.com Next.js clean routes for Privacy, Terms, Data Deletion, and EULA.

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■ Google Firebase & Firestore Architecture

Realtime identity, security rules, and cross-game player data schema under project `webspider-studios-hub`.

Collection / Service	Document Path / Purpose	Key Fields & Data Schema
users Player Profiles	users/{userId} Universal Cross-Game Profile	display_name, username, email, phone, avatar_url, coins, owned_skins, is_premium, daily_streak_count, total_daily_claims, claimed_milestones, webspider_[tier]_coins
transactions Audit Ledger	transactions/{txId} Immutable Coin Log	user_id, type ('credit' 'debit'), amount, reason ('level_complete' 'skin_purchase' 'daily_reward'), created_at
purchases IAP Receipts	purchases/{purchaseId} Google Play Receipt Log	user_id, product_id, purchase_id, transaction_date, verification_data, status, created_at
coupons Promo Codes	coupons/{CODE} Active Rewards	coin_reward, is_active, expiration, created_at
scores Global Leaderboard	scores/{scoreId} Top 100 Realtime Stream	user_id, username, moves, seconds, difficulty, created_at

■ 7-Tier Universal WebSpider Economy Engine

Universal currency values shared across Aqua Sort, Lost Arrow, UniCrush, and upcoming titles.

Tier	Currency Name	Firestore Field	Default Balance	Exchange Rarity
1	Spider Brass	webspider_brass_coins	100	Common Tier
2	Spider Copper	webspider_copper_coins	200	Standard Play
3	Spider Silver	webspider_silver_coins	50	Uncommon
4	Spider Gold	webspider_gold_coins	10	Rare Tier
5	Spider Diamond	webspider_diamond_coins	0	Epic / Competitive
6	Spider Jade	webspider_jade_coins	0	Legendary Quest
7	Spider Obsidian	webspider_obsidian_coins	0	Mythic VIP

■ Meta Developer Setup & Official Compliance

Configuration details for Facebook Login, App ID, Client Token, and Next.js live routes.

Configuration Parameter	Production Value
Meta App ID	1532946758135024
Meta Client Token	95ed14b9943fdac55ccf7ba352da0fbc
Display Name	WebSpider Studios Hub
App Domains	webspiderstudios.com
Privacy Policy URL	https://webspiderstudios.com/privacy
Terms of Service URL	https://webspiderstudios.com/terms
User Data Deletion URL	https://webspiderstudios.com/data-deletion
EULA URL	https://webspiderstudios.com/eula
OAuth Redirect URI	https://webspider-studios-hub.firebaseio.com/__/auth/handler

■ Android Studio & Android CLI Tooling Guide

Standard developer commands to control builds, virtual emulators, and UI layout inspection.

Task / Capability	CLI Command / Workflow
Check Studio Connection	android studio check
Inspect Device Layout	android layout --pretty
Capture Screen to PNG	android screen capture
List Connected Emulators	android emulator list
Search Official Android Docs	android docs search "[query]"
Update Android CLI	android update